

Client Implementation Review

CMPG 325 Milestone 2

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Field	Detail
Student	SEDZE, TANAKA BRUCE
Student number	52827097
Project ID	CMPG325-2026-121
Client ID	CLI-121
Client	Platinum Sound Recording Studio (Rustenburg), Entertainment
Milestone	2. Client Implementation Review due 02 October 2026
Packet Tracer file	52827097_SEDZE_CMPG325-2026-121.pkt
GitHub repository	wantedProgrammer/Computer-Networks

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1. Implementation summary

The Milestone 1 design was built in full in Cisco Packet Tracer and every functional test passed. The network uses the approved 192.168.52.0/24 VLSM plan with no changes to subnets, VLANs or gateways.

What was implemented:

- Five VLANs 10 Admin/Management, 20 Studio, 30 Production, 40 Servers, 50 Guest on three access switches, trunked to a core Layer 3 switch.
- Inter-VLAN routing on the core switch using SVIs plus DHCP pools for VLANs 10, 20, 30 and 50.
- An edge router with a /30 link to the core, a summary static route and NAT overload (PAT) to a simulated ISP.
- A centralized file server in VLAN 40 reachable over FTP.
- CR-14: a WPA2 contractor wireless network in VLAN 50, isolated from internal VLANs by an inbound extended ACL.
- Assigned feature: SSHv2 on all five infrastructure devices, Telnet disabled, VTY access restricted to VLAN 10.

One addition to the Milestone 1 scope: SSH was also applied to the three access switches AND not only the router and core. This keeps the build consistent with the stated requirement that no cleartext protocols run on infrastructure devices.

2. Devices and cabling

Sixteen devices were placed. The ISP router and internet server are a test harness only, used to prove internet access. They sit outside the client's scope.

Device	Packet Tracer model	Role	Management IP
EDGE-R1	2911 router	Edge/WAN gateway, NAT	192.168.52.129
CORE-SW	3560-24PS multilayer	Inter-VLAN routing, DHCP, guest ACL	192.168.52.1
SW-ADMIN	2960-24TT	Access switch, VLAN 10	192.168.52.2
SW-STUDIO	2960-24TT	Access switch, VLAN 20	192.168.52.3
SW-PROD	2960-24TT	Access switch, VLAN 30	192.168.52.4
FILE-SRV	Server-PT	Session file storage (FTP), VLAN 40	192.168.52.98 (static)
GUEST-AP	AccessPoint-PT	Contractor SSID PSR-Contractors, VLAN 50	–
ADMIN-PC1/2, STUDIO-PC1/2, PROD-PC1/2	PC-PT	Staff workstations	DHCP
CONTRACTOR-1	Smartphone-PT	Contractor device (wireless)	DHCP
ISP	2911 router	Simulated ISP (outside scope)	203.0.113.1
INTERNET-SRV	Server-PT	Simulated web/DNS host (outside scope)	198.51.100.10

All wired links use copper straight-through cables.

From	To	Link type
EDGE-R1 G0/0	CORE-SW G0/1	Routed /30 (192.168.52.128/30)
EDGE-R1 G0/1	ISP G0/0	Public /30 (203.0.113.0/30)
ISP G0/1	INTERNET-SRV Fa0	198.51.100.0/24
CORE-SW Fa0/1	SW-ADMIN G0/1	802.1Q trunk, VLAN 10
CORE-SW Fa0/2	SW-STUDIO G0/1	802.1Q trunk, VLANs 10, 20
CORE-SW Fa0/3	SW-PROD G0/1	802.1Q trunk, VLANs 10, 30
CORE-SW Fa0/10	FILE-SRV Fa0	Access, VLAN 40
CORE-SW Fa0/20	GUEST-AP Port 0	Access, VLAN 50
PCs Fa0	Access switch Fa0/1–0/2	Access, department VLAN

3. Access layer: VLANs and trunks

All three access switches were configured without errors and their management SVIs came up in VLAN 10. Each switch carries only its own department VLAN plus the management VLAN, which keeps broadcast domains separate and limits what a mis-patched port can reach.

Switch	Access VLAN (Fa0/1–20)	Trunk G0/1 allowed VLANs	Management SVI
SW-ADMIN	10	10	192.168.52.2/27
SW-STUDIO	20	10, 20	192.168.52.3/27
SW-PROD	30	10, 30	192.168.52.4/27

Design decisions applied on every access switch:

- Unused ports Fa0/21–24 are administratively shut down, so nobody can plug in an unauthorised device.
- PortFast is enabled on user ports, so PCs get a link immediately instead of waiting for spanning tree.
- The trunk allowed list is pruned to the required VLANs only, rather than carrying all VLANs.
- `ip default-gateway 192.168.52.1` lets each switch be managed remotely from VLAN 10.
-

SW-PROD configuration (SW-ADMIN and SW-STUDIO follow the same pattern):

```
hostname SW-PROD
vlan 10
  name ADMIN
vlan 30
  name PRODUCTION
interface range fa0/1 - 20
  switchport mode access
  switchport access vlan 30
  spanning-tree portfast
interface range fa0/21 - 24
  shutdown
interface g0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,30
interface vlan 10
  ip address 192.168.52.4 255.255.255.224
  ip default-gateway 192.168.52.1
```

```

Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/20 but will only
have effect when the interface is in a non-trunking mode.
SW-PROD(config-if-range)#exit
SW-PROD(config)#interface range fa0/21 - 24
SW-PROD(config-if-range)# shutdown

%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
SW-PROD(config-if-range)#exit
SW-PROD(config)#interface g0/1
SW-PROD(config-if)# switchport mode trunk

SW-PROD(config-if)# switchport trunk allowed vlan 10,30
SW-PROD(config-if)#exit
SW-PROD(config)#interface vlan 10
SW-PROD(config-if)# ip address 192.168.52.4 255.255.255.224
SW-PROD(config-if)# no shutdown
SW-PROD(config-if)#exit
SW-PROD(config)#ip default-gateway 192.168.52.1
SW-PROD(config)#end
SW-PROD#write memory
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

```

Evidence 1: SW-PROD configuration accepted with no errors. G0/1 trunk and Vlan10 come up.

4. Core switch: inter-VLAN routing and DHCP

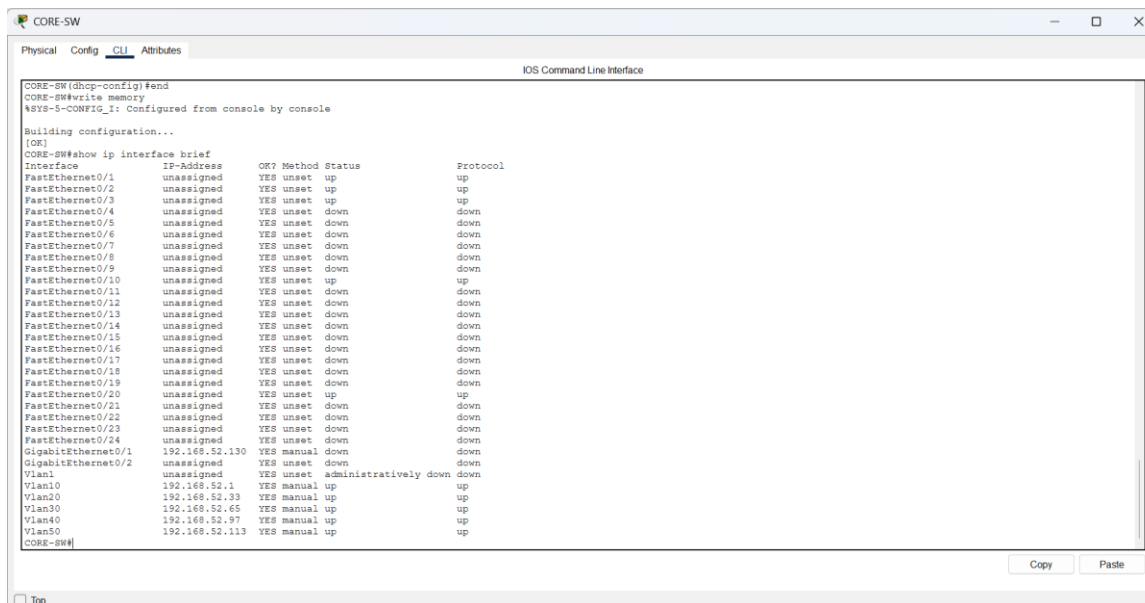
All five SVIs on CORE-SW are up/up with the gateway addresses from the Milestone 1 plan, and every PC received the correct address by DHCP. A ping from Admin to Studio returned TTL=127, confirming one routed hop through the core switch.

VLAN	SVI (gateway)	DHCP pool	Excluded (static) range	First address issued
10 Admin/Mgmt	192.168.52.1/27	ADMIN	.1-.9 (gateway, switch SVIs)	.10
20 Studio	192.168.52.33/27	STUDIO	.33-.37	.38
30 Production	192.168.52.65/27	PRODUCTION	.65-.69	.70
40 Servers	192.168.52.97/28	none (static servers)	-	-
50 Guest	192.168.52.113/28	GUEST	.113-.115	.116

Key configuration:

```
ip routing
interface range fa0/1 - 3
  switchport trunk encapsulation dot1q
  switchport mode trunk
interface g0/1
  no switchport
  ip address 192.168.52.130 255.255.255.252
interface vlan 20
  ip address 192.168.52.33 255.255.255.224
ip route 0.0.0.0 0.0.0.0 192.168.52.129
ip dhcp excluded-address 192.168.52.33 192.168.52.37
ip dhcp pool STUDIO
  network 192.168.52.32 255.255.255.224
  default-router 192.168.52.33
  dns-server 198.51.100.10
```

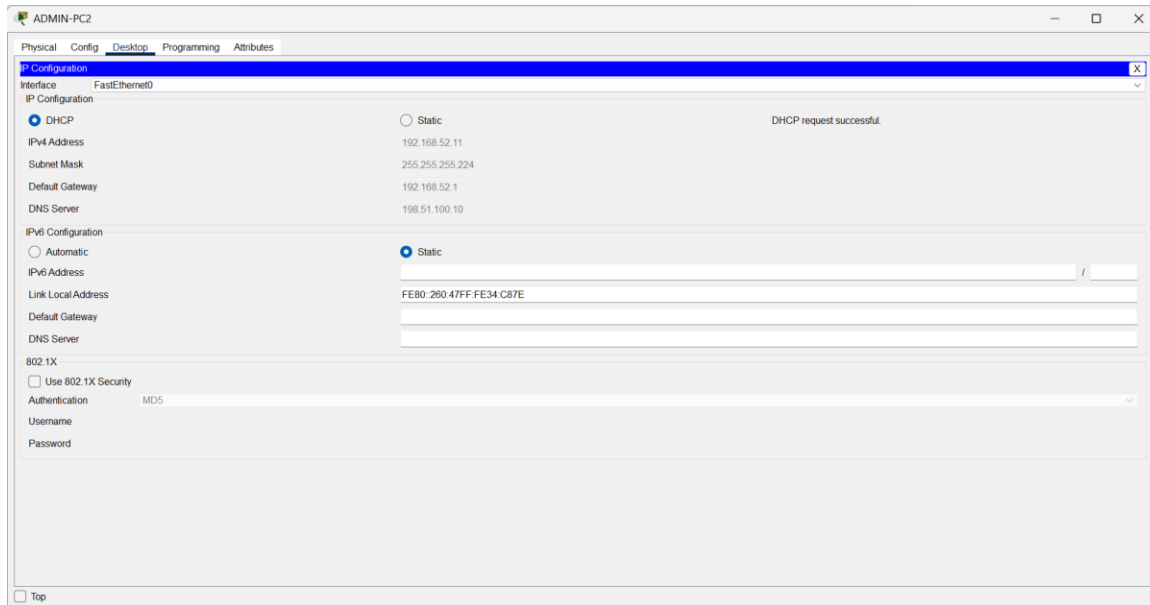
The G0/1 link to the router is a routed port (no switchport), which matches the /30 point-to-point design. The DHCP exclusions keep gateways and switch management addresses out of the pools so they can never be handed to a PC.



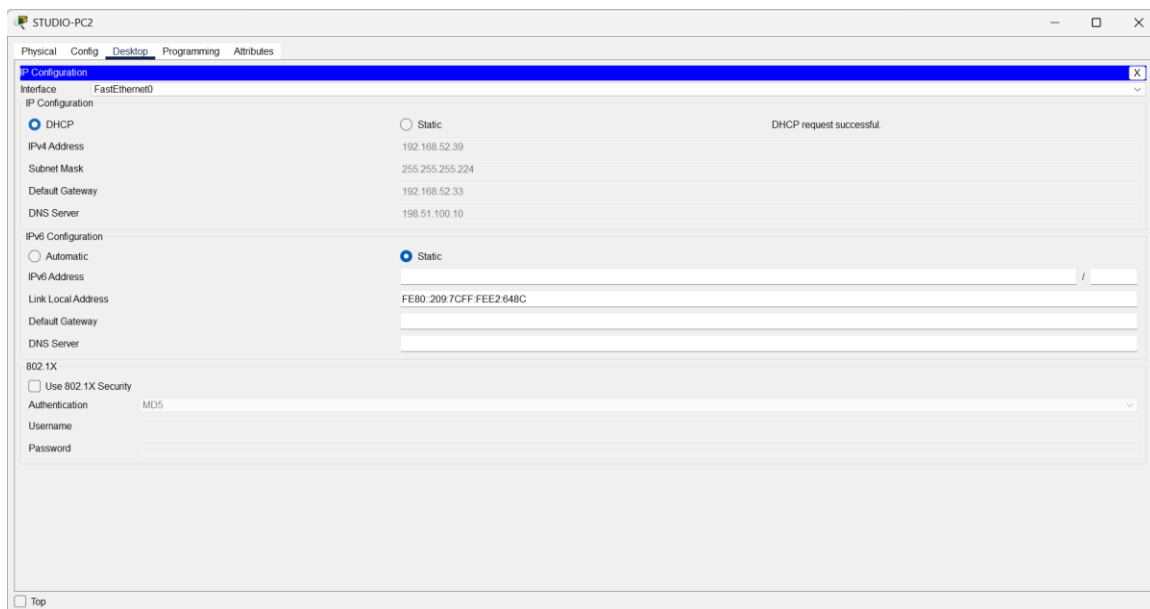
```
CORE-SW(dhcp-config)#end
CORE-SW#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
CORE-SW#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES unset  up          up
FastEthernet0/2 unassigned      YES unset  up          up
FastEthernet0/3 unassigned      YES unset  up          up
FastEthernet0/4 unassigned      YES unset  down        down
FastEthernet0/5 unassigned      YES unset  down        down
FastEthernet0/6 unassigned      YES unset  down        down
FastEthernet0/7 unassigned      YES unset  down        down
FastEthernet0/8 unassigned      YES unset  down        down
FastEthernet0/9 unassigned      YES unset  down        down
FastEthernet0/10 unassigned      YES unset  up          up
FastEthernet0/11 unassigned      YES unset  down        down
FastEthernet0/12 unassigned      YES unset  down        down
FastEthernet0/13 unassigned      YES unset  down        down
FastEthernet0/14 unassigned      YES unset  down        down
FastEthernet0/15 unassigned      YES unset  down        down
FastEthernet0/16 unassigned      YES unset  down        down
FastEthernet0/17 unassigned      YES unset  down        down
FastEthernet0/18 unassigned      YES unset  down        down
FastEthernet0/19 unassigned      YES unset  down        down
FastEthernet0/20 unassigned      YES unset  up          up
FastEthernet0/21 unassigned      YES unset  down        down
FastEthernet0/22 unassigned      YES unset  down        down
FastEthernet0/23 unassigned      YES unset  down        down
FastEthernet0/24 unassigned      YES unset  down        down
GigabitEthernet0/1 192.168.52.130 YES manual down        down
GigabitEthernet0/2 unassigned      YES unset  down        down
Vlan1            unassigned      YES unset  administratively down down
Vlan10           192.168.52.1   YES manual up          up
Vlan20           192.168.52.33  YES manual up          up
Vlan30           192.168.52.65  YES manual up          up
Vlan40           192.168.52.97  YES manual up          up
Vlan50           192.168.52.113 YES manual up          up
CORE-SW#
```

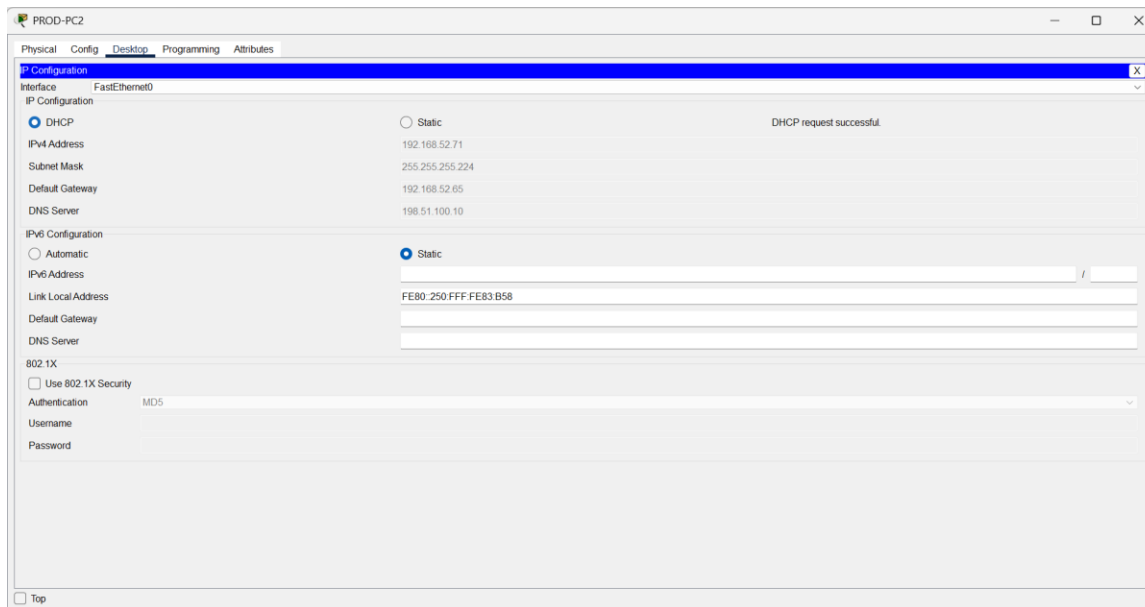
Evidence 2: showS ip interface brief on CORE-SW. Vlan10–50 are up/up with the planned gateway IPs. G0/1 shows down here because the router had not been configured yet.



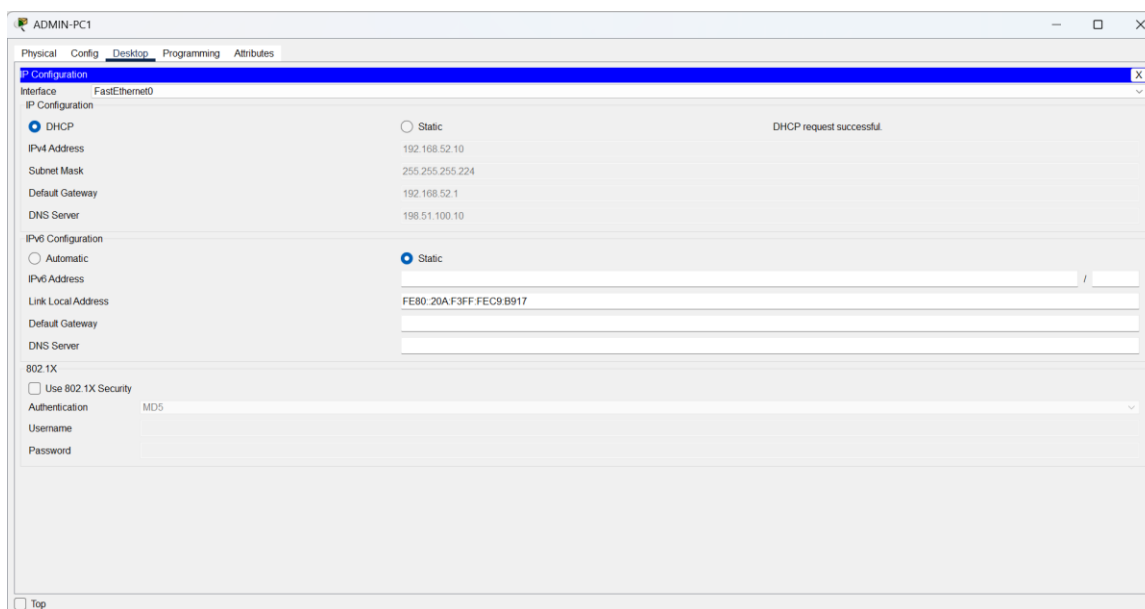
Evidence 3a: ADMIN-PC2 received 192.168.52.11, gateway 192.168.52.1.



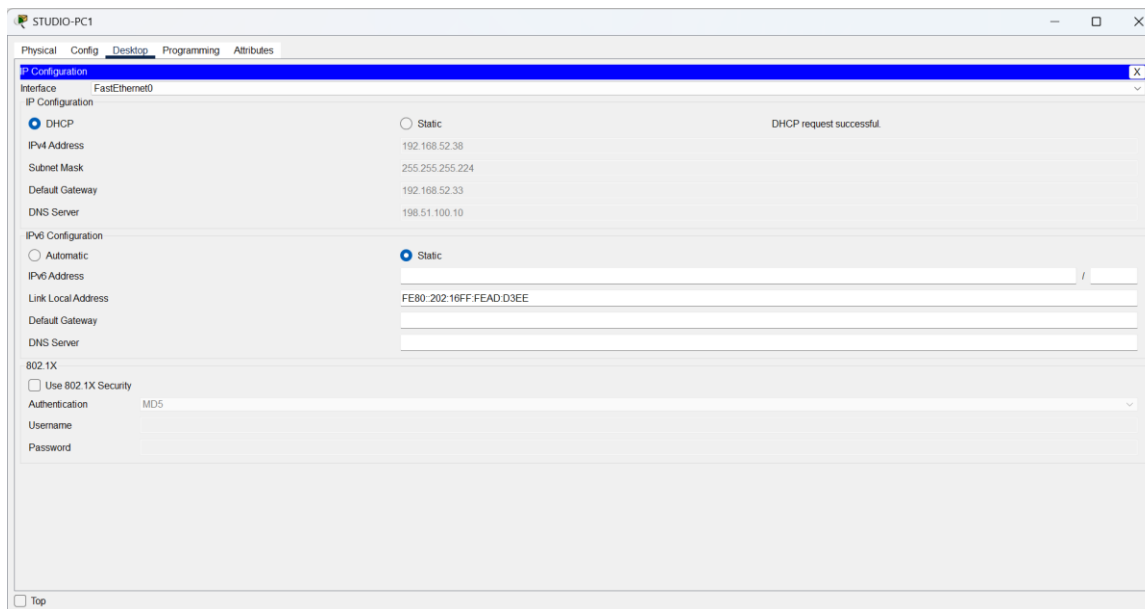
Evidence 3b: STUDIO-PC2 received 192.168.52.39, gateway 192.168.52.33.



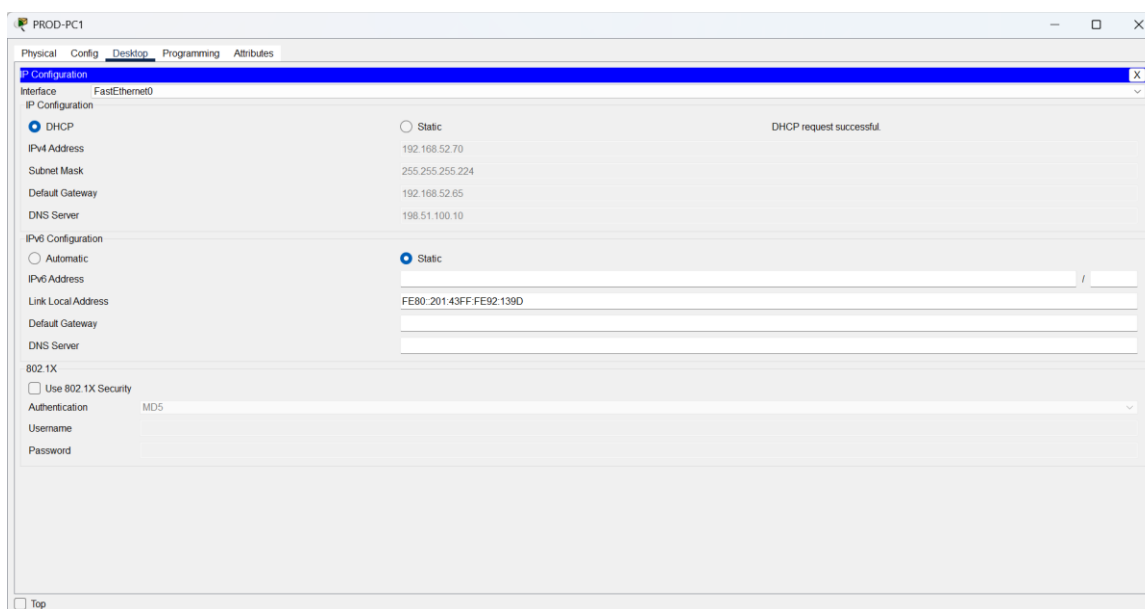
Evidence 3c: PROD-PC2 received 192.168.52.71, gateway 192.168.52.65.



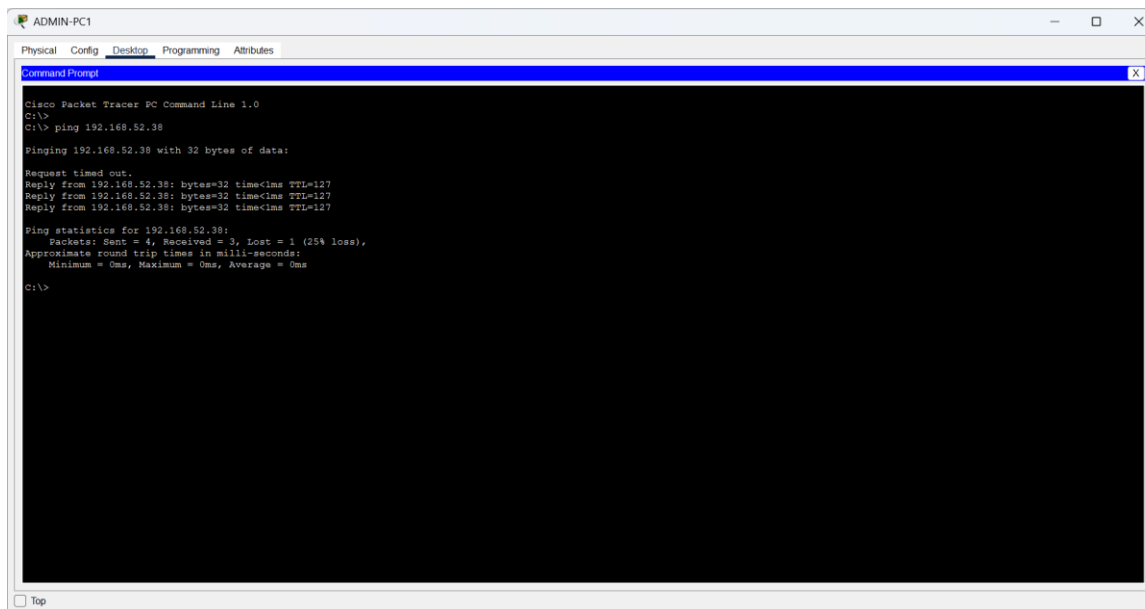
Evidence 3d: ADMIN-PC1 received 192.168.52.10, the first address after the excluded range.



Evidence 3e: STUDIO-PC1 received 192.168.52.38, gateway 192.168.52.33.



Evidence 3f: PROD-PC1 received 192.168.52.70, gateway 192.168.52.65.



Evidence 4: ADMIN-PC1 (VLAN 10) pings 192.168.52.38 (VLAN 20). The first request timed out while ARP resolved. The remaining three replied with TTL=127.

5. Edge router, NAT and internet access

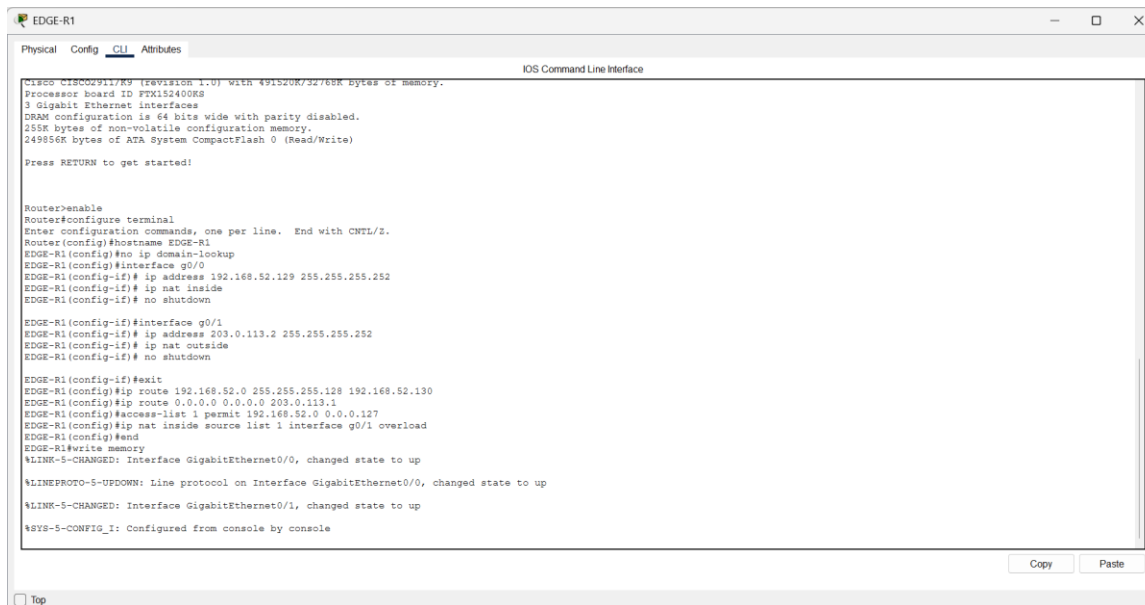
Internal hosts reach the simulated internet through EDGE-R1 using NAT overload (PAT). ADMIN-PC1 pinged 198.51.100.10 and loaded www.example.com, and the router's NAT table shows 192.168.52.10 translated to 203.0.113.2.

Key configuration on EDGE-R1:

```
interface g0/0
ip address 192.168.52.129 255.255.255.252
ip nat inside
interface g0/1
ip address 203.0.113.2 255.255.255.252
ip nat outside
ip route 192.168.52.0 255.255.255.128 192.168.52.130
ip route 0.0.0.0 0.0.0.0 203.0.113.1
access-list 1 permit 192.168.52.0 0.0.0.127
ip nat inside source list 1 interface g0/1 overload
```

Design decisions:

- One summary route (192.168.52.0/25) covers all five VLANs (.0–.127) instead of five separate routes.
- The router keeps only the edge role (routing to the WAN and NAT), in line with the Milestone 1 rationale of a small attack surface.
- PAT lets every internal host share the single public address 203.0.113.2, told apart by source port.
- The ISP router (203.0.113.1, 198.51.100.1) and INTERNET-SRV (198.51.100.10, DNS record www.example.com) are a test harness outside the client's scope. They use RFC 5737 documentation address ranges.



```
EDGE-R1
Physical Config CLI Attributes
IOS Command Line Interface

Cisco C1840-K9-17ES (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400K8
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname EDGE-R1
EDGE-R1(config)#no ip domain-lookup
EDGE-R1(config)#interface g0/0
EDGE-R1(config-if)# ip address 192.168.52.129 255.255.255.252
EDGE-R1(config-if)# ip nat inside
EDGE-R1(config-if)# no shutdown

EDGE-R1(config-if)#interface g0/1
EDGE-R1(config-if)# ip address 203.0.113.2 255.255.255.252
EDGE-R1(config-if)# ip nat outside
EDGE-R1(config-if)# no shutdown

EDGE-R1(config-if)#exit
EDGE-R1(config)#ip route 192.168.52.0 255.255.255.128 192.168.52.130
EDGE-R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.1
EDGE-R1(config)#access-list 1 permit 192.168.52.0 0.0.0.127
EDGE-R1(config)#ip nat inside source list 1 interface g0/1 overload
EDGE-R1(config)#end
EDGE-R1#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Copy Paste
```

Evidence 5: EDGE-R1 configuration accepted; G0/0 to the core switch comes up.

The screenshot shows the IOS Command Line Interface for a simulated ISP router. The interface has tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the following text:

```
http://www.cisco.com/wvl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/R9 (revision 1.0) with 491520R/32768R bytes of memory.
Processor board ID FTX152400R8
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP
ISP(config)#no ip domain-lookup
ISP(config)#interface g0/0
ISP(config-if)# ip address 203.0.113.1 255.255.255.252
ISP(config-if)# no shutdown

ISP(config-if)#interface g0/1
ISP(config-if)# ip address 198.51.100.1 255.255.255.0
ISP(config-if)# no shutdown

ISP(config-if)#end
ISP#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console
```

At the bottom of the window, there are buttons for 'Copy' and 'Paste', and a 'Top' link.

Evidence 6: Simulated ISP router configured; both links up/up.

The screenshot shows the Command Prompt window on ADMIN-PC1. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying the following text:

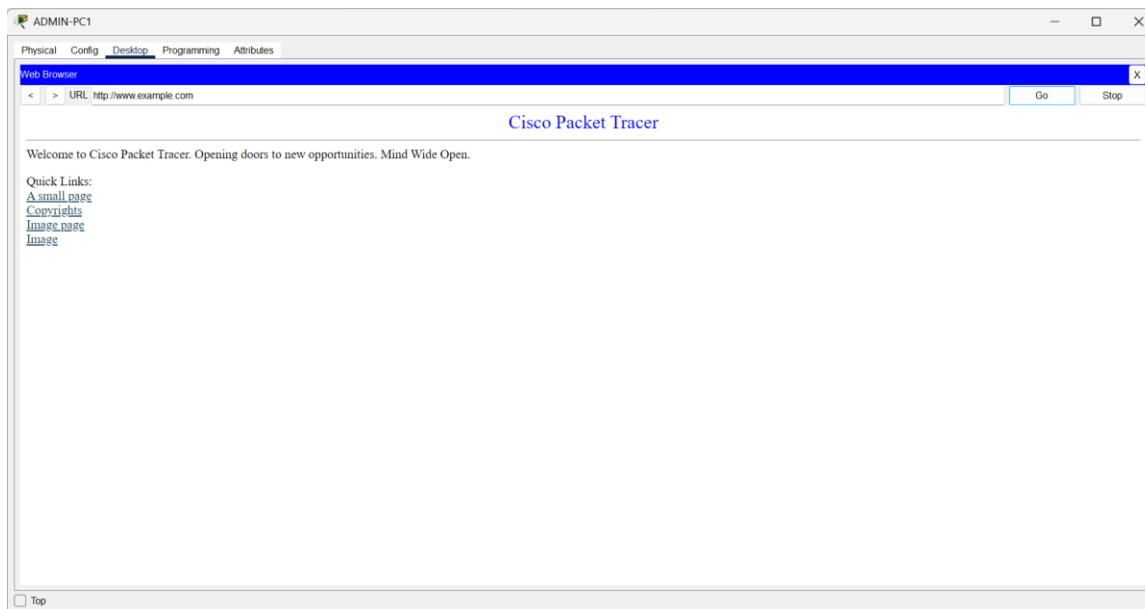
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.51.100.10

Pinging 198.51.100.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Reply from 198.51.100.10: bytes=32 time=125ms TTL=125

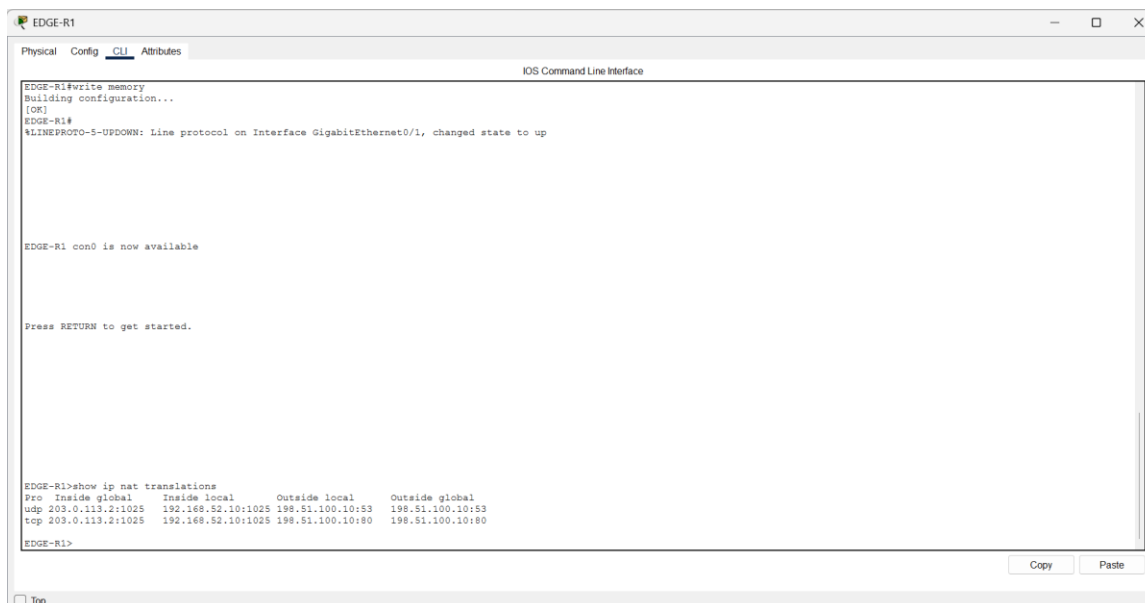
Ping statistics for 198.51.100.10:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

At the bottom of the window, there is a 'Top' link.

Evidence 7: ADMIN-PC1 reaches 198.51.100.10. TTL=125 shows three routed hops (core, edge, ISP). Early timeouts are ARP resolving at each hop.



Evidence 8: *www.example.com loads in ADMIN-PC1's browser, proving DNS resolution and HTTP.*

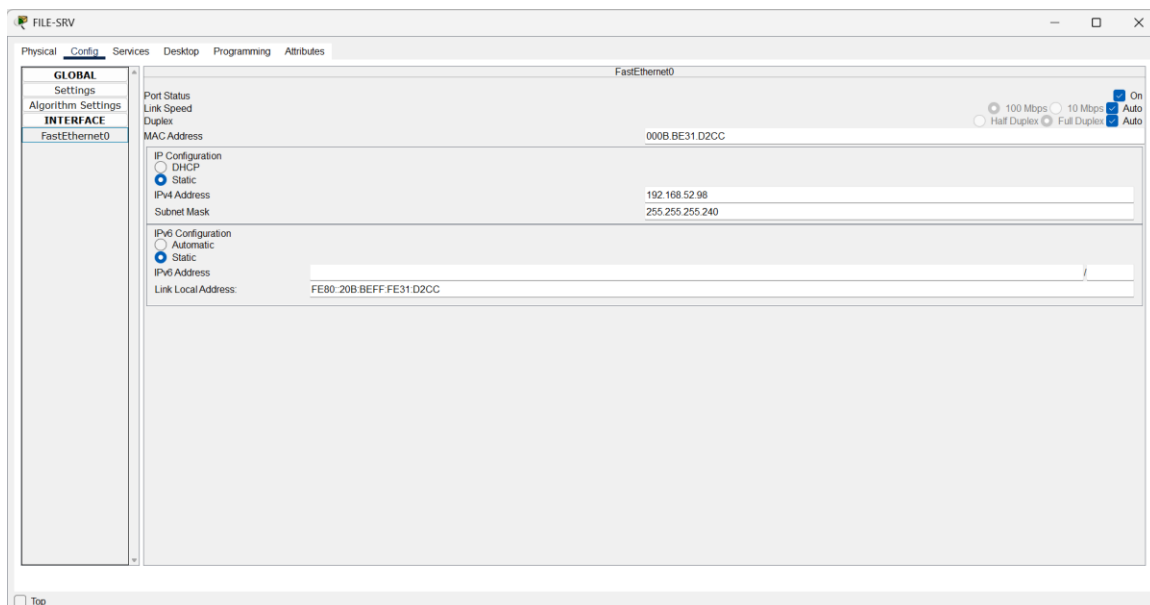


Evidence 9: *show ip nat translations shows the DNS query (UDP 53) and web request (TCP 80) from 192.168.52.10 translated to 203.0.113.2.*

6. File server and centralized storage

PROD-PC2 in the Production VLAN reached FILE-SRV in the Server VLAN and logged in over FTP, meeting the centralized session-file requirement. The server uses a static address because clients must always find it at the same IP.

Setting	Value
IP address / mask	192.168.52.98 / 255.255.255.240
Default gateway	192.168.52.97 (Vlan40 SVI)
DNS	198.51.100.10
Service tested	FTP (Packet Tracer default account)



Evidence 10: FILE-SRV FastEthernet0 set to 192.168.52.98/28, port on.

```

C:\>ping 192.168.52.98

Pinging 192.168.52.98 with 32 bytes of data:

Request timed out.
Reply from 192.168.52.98: bytes=32 time<1ms TTL=127
Reply from 192.168.52.98: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.52.98:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ftp 192.168.52.98
Trying to connect...192.168.52.98
Connected to 192.168.52.98
230- Welcome to FT Ftp server
Username:cisco
331- Username ok, need password
Password:****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.52.98:
 0 : asa942-k8.bin                               5571584
 1 : asa923-k8.bin                               30468096
 2 : c1841-advipservicesk9-mz.124-15.T1.bin      33591768
 3 : c1841-ibase-mz.123-14.T7.bin               13832032
 4 : c1841-ipsasek9-mz.124-12.bin               16599160
 5 : c1900-universalk9-mz.SPA.155-3.M4a.bin      33591768
 6 : c2600-advipservicesk9-mz.124-15.T1.bin      33591768
 7 : c2600-l-mz.122-28.bin                       5571584
 8 : c2600-ipsasek9-mz.124-8.bin                 12169700
 9 : c2800mm-advipservicesk9-mz.124-15.T1.bin     50938004
10 : c2800mm-advipservicesk9-mz.151-4.M4a.bin     33591768
11 : c2800mm-ibase-mz.123-14.T7.bin              5571584
12 : c2800mm-ipsasek9-mz.124-8.bin               16522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin      33591768
14 : c2950-16q412-mz.121-22.EA4.bin             3058048
  
```

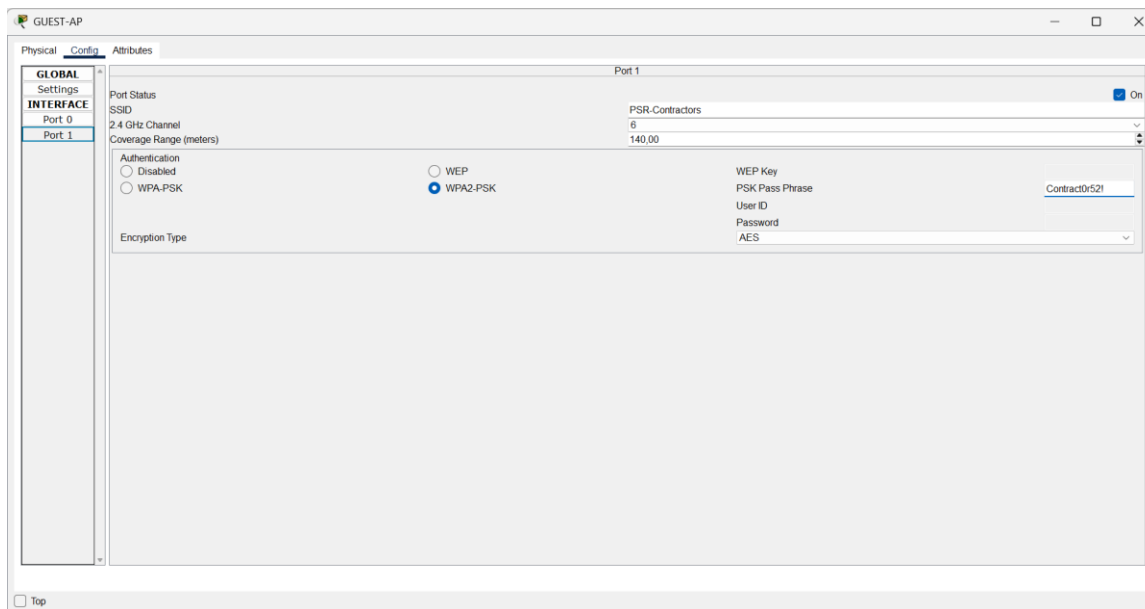
Evidence 11: PROD-PC2 pings 192.168.52.98 (TTL=127, one routed hop) and logs in over FTP (230 Logged in). dir lists the server's directory. The files listed are Packet Tracer's built-in IOS images.

7. CR-14: contractor wireless access

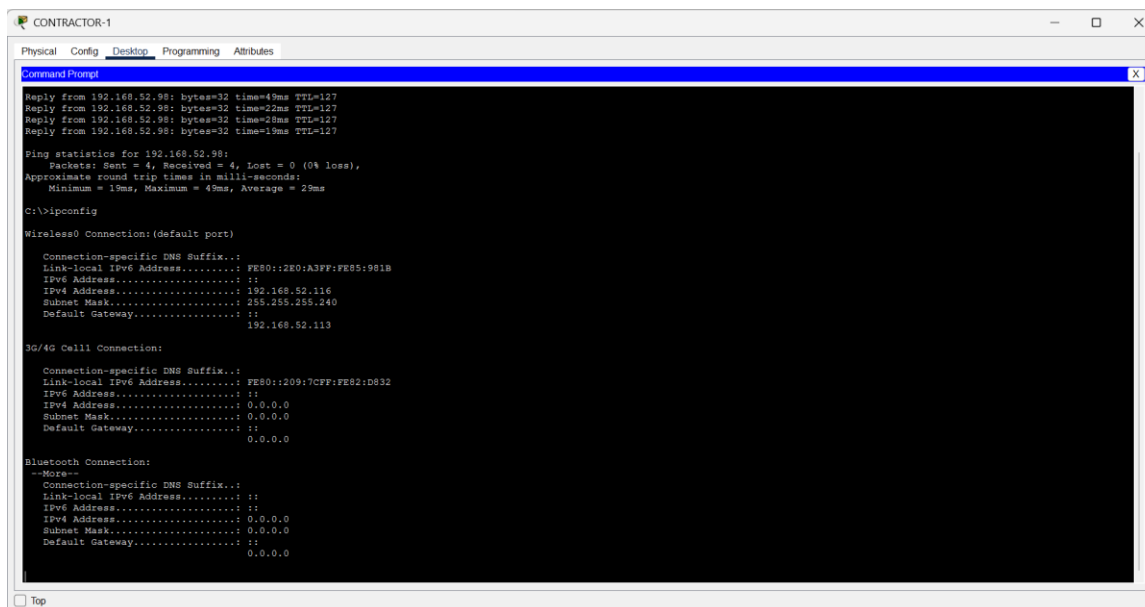
Contractor devices now reach the internet but not the studio's internal VLANs. Before the ACL, CONTRACTOR-1 pinged the file server 4/4. After the ACL, the deny line recorded 8 matches from blocked internal pings, while DNS, web and internet ping were permitted.

Wireless setup:

Setting	Value
Access point	GUEST-AP on CORE-SW Fa0/20 (access VLAN 50)
SSID	PSR-Contractors
Security	WPA2-PSK, AES
Client	CONTRACTOR-1 (Smartphone-PT), DHCP from pool GUEST (.116 upward)



Evidence 12a: GUEST-AP Port 1 configured with SSID PSR-Contractors, WPA2-PSK authentication and AES encryption.



Evidence 12b: CONTRACTOR-1's Wireless0 interface received 192.168.52.116/28 with gateway 192.168.52.113 from the GUEST pool, confirming it joined VLAN 50.

GUEST-IN ACL, applied inbound on the Vlan50 SVI:

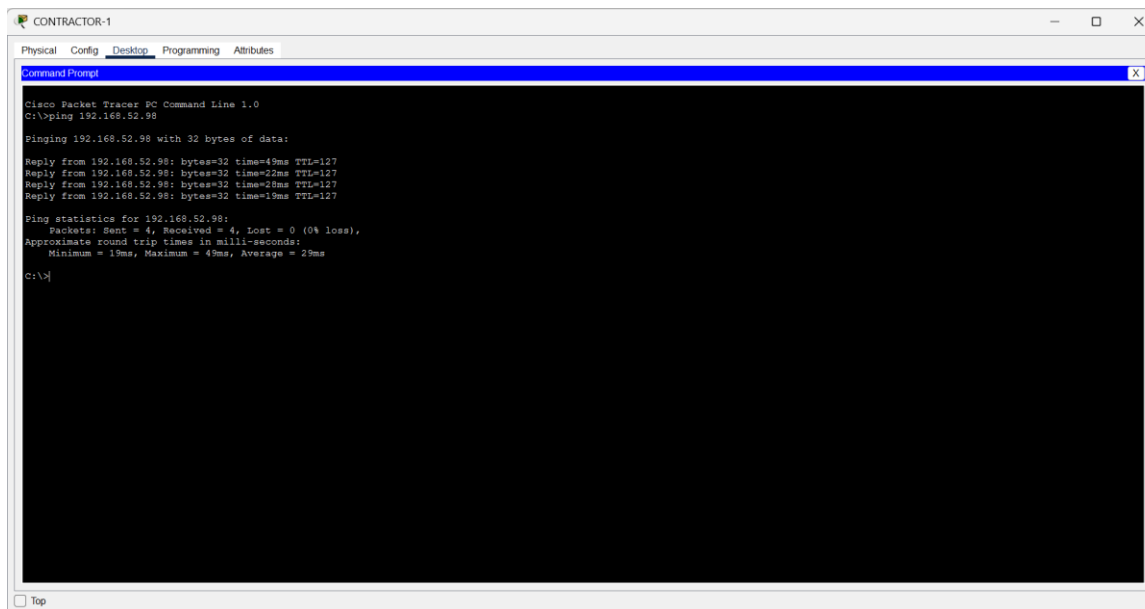
```
ip access-list extended GUEST-IN
remark CR-14 contractor wireless - internet only
permit udp any eq bootpc any eq bootps
deny ip 192.168.52.112 0.0.0.15 192.168.52.0 0.0.0.255
permit udp 192.168.52.112 0.0.0.15 any eq domain
permit tcp 192.168.52.112 0.0.0.15 any eq www
permit tcp 192.168.52.112 0.0.0.15 any eq 443
permit icmp 192.168.52.112 0.0.0.15 any echo
deny ip any any
interface vlan 50
ip access-group GUEST-IN in
```

The order of the entries is deliberate:

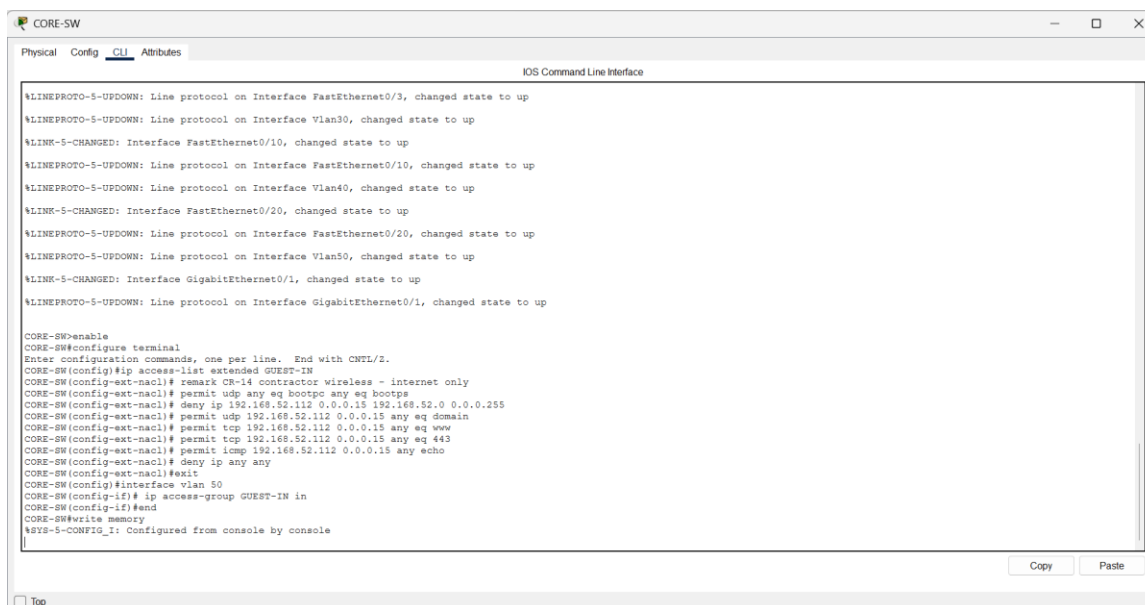
1. DHCP is permitted first, so a contractor can still obtain an address.
2. Everything to 192.168.52.0/24 is denied next, covering VLANs 10–40, the router link and the gateway itself.
3. Only after that are DNS, HTTP, HTTPS and outbound ping allowed, so they can only ever reach external addresses.
4. The explicit final deny gives a hit counter for anything else.

ACL results from show access-lists after testing:

Line	Entry	Matches	What it proves
20	deny to 192.168.52.0/24	8	Two internal pings (4 packets each) were dropped
30	permit DNS	1	www.example.com was resolved
40	permit HTTP	5	The web page loaded
60	permit ICMP echo	4	Ping to the internet server was allowed
10	permit DHCP	0	Expected: the device leased its address before the ACL was applied



Evidence 12c (before): CONTRACTOR-1 pings FILE-SRV 192.168.52.98 successfully (0% loss). This is the gap CR-14 requires closing.



Evidence 13: GUEST-IN entered with no errors and applied inbound on interface Vlan50.

CONTRACTOR-1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>
C:\>cls
Invalid Command.

C:\>ping 192.168.52.98

Pinging 192.168.52.98 with 32 bytes of data:

Reply from 192.168.52.113: Destination host unreachable.
Reply from 192.168.52.113: Destination host unreachable.
Reply from 192.168.52.113: Destination host unreachable.
Reply from 192.168.52.113: Destination host unreachable.

Ping statistics for 192.168.52.98:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.52.10

Pinging 192.168.52.10 with 32 bytes of data:

Reply from 192.168.52.113: Destination host unreachable.
Reply from 192.168.52.113: Destination host unreachable.
Reply from 192.168.52.113: Destination host unreachable.
Reply from 192.168.52.113: Destination host unreachable.

Ping statistics for 192.168.52.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 198.51.100.10

Pinging 198.51.100.10 with 32 bytes of data:

Reply from 198.51.100.10: bytes=32 time=39ms TTL=125
Reply from 198.51.100.10: bytes=32 time=20ms TTL=125
Reply from 198.51.100.10: bytes=32 time=14ms TTL=125
Reply from 198.51.100.10: bytes=32 time=20ms TTL=125

Ping statistics for 198.51.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 39ms, Average = 23ms

C:\>
```

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Evidence 14a (after): CONTRACTOR-1 can no longer reach FILE-SRV (192.168.52.98) or ADMIN-PC1 (192.168.52.10). The Vlan50 gateway (192.168.52.113) returns "destination host unreachable" for both. The ping to INTERNET-SRV (198.51.100.10) still succeeds. These 8 dropped and 4 permitted packets are the match counts shown in Evidence 14.

CORE-SW

Physical Config CLI Attributes

IOS Command Line Interface

```
$LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/20, changed state to up
$LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, changed state to up
$LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
$LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

CORE-SW#enable
CORE-SW#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CORE-SW(config)#ip access-list extended GUEST-IN
CORE-SW(config-ext-nacl)# remark CR-14 contractor wireless - internet only
CORE-SW(config-ext-nacl)# permit udp any eq bootpc any eq bootps
CORE-SW(config-ext-nacl)# deny ip 192.168.52.112 0.0.0.15 192.168.52.0 0.0.0.255
CORE-SW(config-ext-nacl)# permit udp 192.168.52.112 0.0.0.15 any eq domain
CORE-SW(config-ext-nacl)# permit tcp 192.168.52.112 0.0.0.15 any eq www
CORE-SW(config-ext-nacl)# permit tcp 192.168.52.112 0.0.0.15 any eq 443
CORE-SW(config-ext-nacl)# permit icmp 192.168.52.112 0.0.0.15 any echo
CORE-SW(config-ext-nacl)# deny ip any any
CORE-SW(config-ext-nacl)# exit
CORE-SW(config)#interface Vlan 50
CORE-SW(config-if)# ip access-group GUEST-IN in
CORE-SW(config-if)#end
CORE-SW#write memory
$SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
CORE-SW# show access-lists
Extended IP access list GUEST-IN
 10 permit udp any eq bootpc any eq bootps
 20 deny ip 192.168.52.112 0.0.0.15 192.168.52.0 0.0.0.255 (8 match(es))
 30 permit udp 192.168.52.112 0.0.0.15 any eq domain (1 match(es))
 40 permit tcp 192.168.52.112 0.0.0.15 any eq www (5 match(es))
 50 permit tcp 192.168.52.112 0.0.0.15 any eq 443
 60 permit icmp 192.168.52.112 0.0.0.15 any echo (4 match(es))
 70 deny ip any any

CORE-SW#
```

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Evidence 14 (after): match counters on GUEST-IN confirm internal traffic is dropped and internet traffic is permitted.

8. Assigned feature: secure management with SSH

SSHv2 is running on all five infrastructure devices, Telnet is disabled, and only VLAN 10 can open a management session. Admin reached CORE-SW, EDGE-R1 and SW-STUDIO over an AES-128 encrypted session, while a Studio PC was refused before any password prompt.

The same block was applied to EDGE-R1, CORE-SW, SW-ADMIN, SW-STUDIO and SW-PROD (passwords redacted):

```
ip domain-name platinumsound.local
username netadmin privilege 15 secret <redacted>
enable secret <redacted>
service password-encryption
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
access-list 10 remark SSH allowed from VLAN 10 only
access-list 10 permit 192.168.52.0 0.0.0.31
line vty 0 15
  transport input ssh
  login local
  access-class 10 in
  exec-timeout 5 0
line console 0
  login local
  exec-timeout 5 0
banner motd # Authorised access only - Platinum Sound Studio #
```

What each control does:

Command	Purpose
crypto key generate rsa ... modulus 1024	Creates the RSA key pair SSH needs; SSHv2 requires at least 768 bits
ip ssh version 2	Disables SSHv1, which has known weaknesses
ip ssh time-out 60, authentication-retries 3	Limits how long and how often a login can be attempted, slowing password guessing
transport input ssh	Only SSH is accepted on the VTY lines, so Telnet (cleartext) is refused
login local + username ... secret	Authenticates against a local account stored as a hash
access-class 10 in	Only 192.168.52.0/27 (VLAN 10) may even attempt a session
exec-timeout 5 0	Idle sessions close after 5 minutes
banner motd	Legal warning shown before login

On EDGE-R1, access list 10 (SSH) is separate from access list 1 (NAT), so the two policies cannot interfere.

```

CORE-SW
Physical Config CLI Attributes
IOS Command Line Interface

CORE-SW(config-ext-nacl)# deny ip 192.168.52.112 0.0.0.15 192.168.52.0 0.0.0.255
CORE-SW(config-ext-nacl)# permit udp 192.168.52.112 0.0.0.15 any eq domain
CORE-SW(config-ext-nacl)# permit tcp 192.168.52.112 0.0.0.15 any eq www
CORE-SW(config-ext-nacl)# permit tcp 192.168.52.112 0.0.0.15 any eq 443
CORE-SW(config-ext-nacl)# permit icmp 192.168.52.112 0.0.0.15 any echo
CORE-SW(config-ext-nacl)# deny ip any any
CORE-SW(config-ext-nacl)# exit
CORE-SW(config)# interface vlan 50
CORE-SW(config-if)# ip access-group GUEST-IN in
CORE-SW(config-if)# end
CORE-SW# write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
CORE-SW# show access-lists
Extended IP access list GUEST-IN
 10 permit udp any eq bootpc any eq bootps
 20 deny ip 192.168.52.112 0.0.0.15 192.168.52.0 0.0.0.255 (8 match(es))
 30 permit udp 192.168.52.112 0.0.0.15 any eq domain (1 match(es))
 40 permit tcp 192.168.52.112 0.0.0.15 any eq www (5 match(es))
 50 permit tcp 192.168.52.112 0.0.0.15 any eq 443
 60 permit icmp 192.168.52.112 0.0.0.15 any echo (4 match(es))
 70 deny ip any any

CORE-SW#enable
CORE-SW#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CORE-SW(config)#ip domain-name platinumsound.local
CORE-SW(config)#username netadmin privilege 15 secret Tank#Admin52
CORE-SW(config)#enable secret Bruce#Plat52
CORE-SW(config)#service password-encryption
CORE-SW(config)#crypto key generate rsa general-keys modulus 1024
The name for the keys will be: CORE-SW.platinumsound.local

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 6:02:1.698: %SSH-5-ENABLED: SSH 1.99 has been enabled
CORE-SW(config)#ip ssh version 2
CORE-SW(config)#ip ssh time-out 60
CORE-SW(config)#ip ssh authentication-retries 3
Copy Paste
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```

Evidence 15: RSA keys generated as CORE-SW.platinumsound.local and SSH enabled on CORE-SW.

The screenshot shows the IOS Command Line Interface for a device named CORE-SW. The user is in configuration mode, and the following commands have been entered: `configure terminal`, `ip domain-name platinumsound.local`, `username netadmin privilege 15 secret Tank$Admin52`, `enable secret Bruce$Plat52`, `service password-encryption`, `crypto key generate rsa general-keys modulus 1024`, and `line vty 0 15`. The output shows that SSH version 2 is enabled, the key modulus size is 1024 bits, and the SSH service is running. The user then enters `transport input ssh`, `login local`, `access-class 10 in`, `exec-timeout 5 0`, and `exit`. The user then enters `line console 0`, `login local`, `exec-timeout 5 0`, and `exit`. The user then enters `banner motd # Authorized access only - Platinum Sound Studio #`, `end`, `write memory`, and `show ip ssh`. The output of `show ip ssh` shows that SSH is enabled, version 2.0, with a timeout of 60 seconds and 3 authentication retries. The user then enters `exit`.

```
CORE-SW#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CORE-SW(config)#ip domain-name platinumsound.local
CORE-SW(config)#username netadmin privilege 15 secret Tank$Admin52
CORE-SW(config)#enable secret Bruce$Plat52
CORE-SW(config)#service password-encryption
CORE-SW(config)#crypto key generate rsa general-keys modulus 1024
The name for the keys will be: CORE-SW.platinumsound.local

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 6:02:1.698: %SSH-5-ENABLED: SSH 1.99 has been enabled
CORE-SW(config)#ip ssh version 2
CORE-SW(config)#ip ssh time-out 60
CORE-SW(config)#ip ssh authentication-retries 3
CORE-SW(config)#access-list 10 remark SSH allowed from VLAN 10 only
CORE-SW(config)#access-list 10 permit 192.168.52.0 0.0.0.31
CORE-SW(config)#line vty 0 15
CORE-SW(config-line)# transport input ssh
CORE-SW(config-line)# login local
CORE-SW(config-line)# access-class 10 in
CORE-SW(config-line)# exec-timeout 5 0
CORE-SW(config-line)#exit
CORE-SW(config)#line console 0
CORE-SW(config-line)# login local
CORE-SW(config-line)# exec-timeout 5 0
CORE-SW(config-line)#exit
CORE-SW(config)#banner motd # Authorized access only - Platinum Sound Studio #
CORE-SW(config)#end
CORE-SW#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
CORE-SW#write memory
Building configuration...
[OK]
CORE-SW#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
CORE-SW#
```

Evidence 16: `show ip ssh` on CORE-SW reports SSH Enabled – version 2.0, 60 s timeout, 3 retries.

The screenshot shows the IOS Command Line Interface for a device named EDGE-R1. The user is in configuration mode, and the following commands have been entered: `enable`, `configure terminal`, `ip domain-name platinumsound.local`, `username netadmin privilege 15 secret Tank$Admin52`, `enable secret Bruce$Plat52`, `service password-encryption`, `crypto key generate rsa general-keys modulus 1024`, and `line vty 0 15`. The output shows that SSH version 2 is enabled, the key modulus size is 1024 bits, and the SSH service is running. The user then enters `transport input ssh`, `login local`, `access-class 10 in`, `exec-timeout 5 0`, and `exit`. The user then enters `line console 0`, `login local`, `exec-timeout 5 0`, and `exit`. The user then enters `banner motd # Authorized access only - Platinum Sound Studio #`, `end`, `write memory`, and `show ip ssh`. The output of `show ip ssh` shows that SSH is enabled, version 2.0, with a timeout of 60 seconds and 3 authentication retries. The user then enters `exit`.

```
EDGE-R1#enable
EDGE-R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
EDGE-R1(config)#ip domain-name platinumsound.local
EDGE-R1(config)#username netadmin privilege 15 secret Tank$Admin52
EDGE-R1(config)#enable secret Bruce$Plat52
EDGE-R1(config)#service password-encryption
EDGE-R1(config)#crypto key generate rsa general-keys modulus 1024
The name for the keys will be: EDGE-R1.platinumsound.local

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 6:11:55.798: %SSH-5-ENABLED: SSH 1.99 has been enabled
EDGE-R1(config)#ip ssh version 2
EDGE-R1(config)#ip ssh time-out 60
EDGE-R1(config)#ip ssh authentication-retries 3
EDGE-R1(config)#access-list 10 remark SSH allowed from VLAN 10 only
EDGE-R1(config)#access-list 10 permit 192.168.52.0 0.0.0.31
EDGE-R1(config)#line vty 0 15
EDGE-R1(config-line)# transport input ssh
EDGE-R1(config-line)# login local
EDGE-R1(config-line)# access-class 10 in
EDGE-R1(config-line)# exec-timeout 5 0
EDGE-R1(config-line)#exit
EDGE-R1(config)#line console 0
EDGE-R1(config-line)# login local
EDGE-R1(config-line)# exec-timeout 5 0
EDGE-R1(config-line)#exit
EDGE-R1(config)#banner motd # Authorized access only - Platinum Sound Studio #
EDGE-R1(config)#end
EDGE-R1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
EDGE-R1#
```

Evidence 17: The same SSH block accepted on EDGE-R1, including line `vty 0 15`. The three access switches were configured identically.

```

ADMIN-PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.51.100.10

Pinging 198.51.100.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Reply from 198.51.100.10: bytes=32 time<1ms TTL=125

Ping statistics for 198.51.100.10:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ssh -l netadmin 192.168.52.1

Password:
    Authorised access only - Platinum Sound Studio

CORE-SW#show ssh
Connection  Version Mode Encryption      Hmac  State      Username
2          1.99  IN   aes128-cbc    hmac-sha1  Session Started  netadmin
2          1.99  OUT  aes128-cbc    hmac-sha1  Session Started  netadmin
No SSHv1 server connections running.
CORE-SW#

```

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Evidence 18: ADMIN-PC1 logs into CORE-SW; show ssh lists the session as aes128-cbc with hmac-sha1 for user netadmin. The session table labels it 1.99, but show ip ssh (Evidence 16) confirms the server runs version 2.0.

```

ADMIN-PC1
Physical Config Desktop Programming Attributes
Command Prompt
Ping statistics for 198.51.100.10:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ssh -l netadmin 192.168.52.1

Password:
    Authorised access only - Platinum Sound Studio

CORE-SW#show ssh
Connection  Version Mode Encryption      Hmac  State      Username
2          1.99  IN   aes128-cbc    hmac-sha1  Session Started  netadmin
2          1.99  OUT  aes128-cbc    hmac-sha1  Session Started  netadmin
No SSHv1 server connections running.
CORE-SW#exit

[Connection to 192.168.52.1 closed by foreign host]
C:\>ssh -l netadmin 192.168.52.129

Password:
    Authorised access only - Platinum Sound Studio

EDGE-R1#exit

[Connection to 192.168.52.129 closed by foreign host]
C:\>ssh -l netadmin 192.168.52.3

Password:
    Authorised access only - Platinum Sound Studio

SW-STUDIO#exit

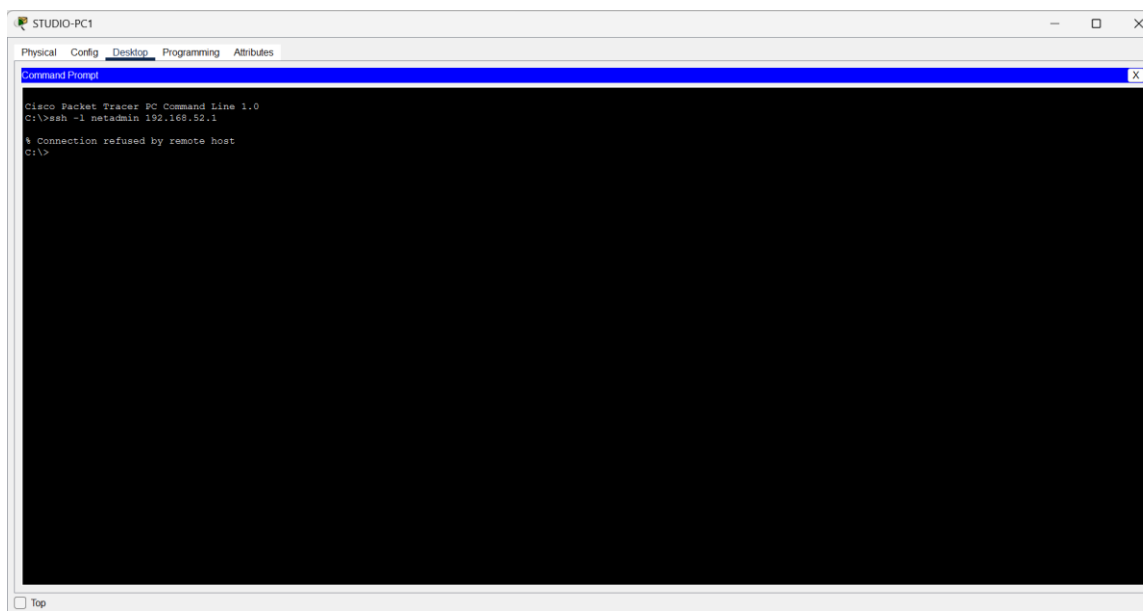
[Connection to 192.168.52.3 closed by foreign host]
C:\>telnet 192.168.52.1
Trying 192.168.52.1 ...Open

[Connection to 192.168.52.1 closed by foreign host]
C:\>

```

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Evidence 19: From ADMIN-PC1, SSH logins succeed to 192.168.52.1, .129 and .3. telnet 192.168.52.1 opens TCP but is closed immediately with no login prompt, because the VTY lines accept SSH only.



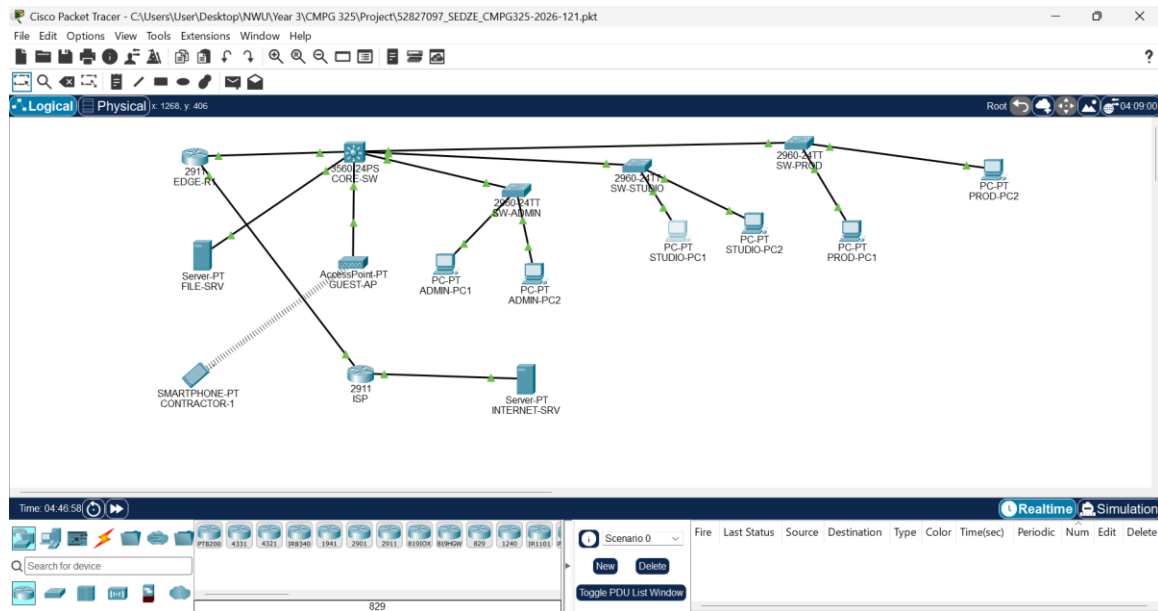
Evidence 20: From STUDIO-PC1 (VLAN 20), ssh -l netadmin 192.168.52.1 returns “Connection refused by remote host” with no password prompt. The access-class blocks the attempt before authentication, as shown in the Milestone 1 SSH flow diagram.

9. Testing summary

All 15 tests produced the expected result.

#	Test	From → To	Expected	Result	Evidence
1	VLAN and trunk config	SW-PROD	Accepted, trunk up	Pass	1
2	SVIs up	CORE-SW	Vlan10–50 up/up	Pass	2
3	DHCP, all user VLANs	PCs → CORE-SW	Correct pool and gateway	Pass	3a–3f
4	Inter-VLAN routing	ADMIN-PC1 → STUDIO PC	Reply, TTL=127	Pass	4
5	Internet reachability	ADMIN-PC1 → 198.51.100.10	Reply, TTL=125	Pass	7
6	DNS and web	ADMIN-PC1 →	Page loads	Pass	8

#	Test	From → To	Expected	Result	Evidence
		www.example.com			
7	NAT overload	EDGE-R1	.10 translated to 203.0.113.2	Pass	9
8	File server access	PROD-PC2 → FILE-SRV	Ping and FTP login	Pass	11
9	Guest open before ACL	CONTRACT OR-1 → FILE-SRV	Reply (baseline)	Pass	12c
10	Guest isolation after ACL	CONTRACT OR-1 → internal	Dropped (8 matches)	Pass	14a, 14
11	Guest internet after ACL	CONTRACT OR-1 → internet	DNS, HTTP, ping permitted	Pass	14a, 14
12	SSHv2 enabled	CORE-SW	version 2.0	Pass	15, 16
13	SSH from VLAN 10	ADMIN-PC1 → .1, .129, .3	Encrypted login	Pass	18, 19
14	Telnet blocked	ADMIN-PC1 → .1	No login prompt	Pass	19
15	SSH from other VLANs blocked	STUDIO-PC1 → .1	Connection refused	Pass	20



Evidence 21: Full logical topology in Packet Tracer, with all links up (green).

10. Deviations, limitations and issues resolved

The build matches the Milestone 1 design with one scope addition and one simulator limitation.

Item	Type	Detail and handling
SSH on access switches	Addition	Milestone 1 named only the router and core. Access switches were added so no infrastructure device accepts cleartext management.
Time-based ACL (CR-14 "after hours")	Simulator limitation	Packet Tracer does not support time-range. Isolation is demonstrated in the simulation; the time restriction is documented below as real IOS configuration.
Simulated ISP and internet server	Test harness	Added only to prove internet access and NAT; outside the client scope.
Router froze after a mistyped command	Resolved	Before no ip domain-lookup was set, a stray word triggered a DNS lookup. The unconfigured router was power-cycled and configured cleanly.
SW-STUDIO initially saved a blank config	Resolved	The Config tab's Save button ran before the commands were pasted; the configuration was then applied and saved.

Real-IOS time restriction for CR-14 (not supported in Packet Tracer):

```
time-range CONTRACTOR-HOURS
periodic daily 18:00 to 23:59
periodic daily 0:00 to 6:00
!
ip access-list extended GUEST-IN
permit tcp 192.168.52.112 0.0.0.15 any eq www time-range CONTRACTOR-HOURS
permit tcp 192.168.52.112 0.0.0.15 any eq 443 time-range CONTRACTOR-HOURS
```

The window is split into two periodic entries because a single entry does not wrap past midnight.

11. GitHub portfolio and next steps

The repository [wantedProgrammer/Computer-Networks](#) is updated for Milestone 2 with the working .pkt file, device configurations and all testing screenshots.